

David R. Miller

PHD CANDIDATE - UNIVERSITY OF BRITISH COLUMBIA

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Education

University of British Columbia

Vancouver, BC, Canada

PHD, ASTRONOMY

2022–Present

Supervisors: Dr. Jeremy Heyl and Dr. Harvey Richer (1944–2023)

Thesis: Probing Stellar Evolution With White Dwarfs in Open Clusters

Expected Completion: Spring 2027

University of British Columbia

Vancouver, BC, Canada

MSc, ASTRONOMY

2020–2022

Supervisor: Dr. Harvey Richer (1944–2023)

Thesis: [The White Dwarf Initial-Final Mass Relation: Ultramassive White Dwarfs Ejected from Young Open Star Clusters](#)

Simon Fraser University

Burnaby, BC, Canada

BSc, PHYSICS HONOURS (WITH DISTINCTION)

2016–2020

Supervisor: Dr. Levon Pogorian

Thesis: [Dark Sector Cosmic Fluid Interactions as a Solution to the Observed Disagreement in Measurements of the Hubble Expansion Rate](#)

Awards

UBC Four Year Fellowship (4YF), \$98,000

2023–2027

President's Academic Excellence Initiative PhD Award, \$4,800

2022–2026

Dunlap Institute Summer Undergraduate Research Program Fellowship (SURP), \$9,500

2019

NSERC Undergraduate Student Research Award (USRA), \$5,250

2018

Ken Caple College Transfer Entrance Scholarship, \$3,500

2016

Publications

First Author (4)

1. **Miller, D. R.**, Heyl, J., and Tremblay, P-E. Investigating the Observational Progenitor Mass Gap in the White Dwarf Initial-Final Mass Relation. I. Cluster Census and Characterization of the First White Dwarfs in the Gap. 2026, *ApJ*, accepted.
2. **Miller, D. R.**, Caiazzo, I., Heyl, J., et al. The White Dwarf Initial-Final Mass Relation from Open Clusters in Gaia DR3. 2026, *ApJ*, 996:69.
3. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. An Extremely Massive White Dwarf Escaped From the Hyades Star Cluster. 2023, *ApJL*, 956:L41. **Featured in:** AAS Nova, Space.com, Universe Today.
4. **Miller, D. R.**, Caiazzo, I., Heyl, J. et al. The Ultramassive White Dwarfs of the Alpha Persei Cluster. 2022, *ApJL*, 926:L24.

Second/Third Author (3)

1. Jadhav, V. V., Kroupa, P., **Miller, D. R.**, Sahu, S., and Franktisek, D. The white dwarf population of open

clusters and their tidal tails: Tracers of contamination and stellar interactions. 2026, *A&A*, accepted.

2. Yan, H., **Miller, D. R.**, Zhao, J., Guo, J., and Li, C. Escaped White Dwarf Candidates from Open Clusters. 2026, *ApJS*, 286:5.
3. Yan, H., Wang, L., **Miller, D. R.**, et al. Searching for White Dwarf Candidates Formed Through Binary Evolution in Star Clusters. 2026, *ApJ*, 999:167.

Contributing Author (4)

1. Cristea, A. A., Caiazzo, I., Cunningham, T., Raymond, J. C., Vennes, S., Kawka, A., Desai, A., **Miller, D. R.**, et al. A half ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant: A new class of white dwarf merger remnants with X-ray emission. 2026, *A&A*, 706, A188.
2. Côté, P., et al. (including **Miller, D. R.**). The CASTOR mission. 2025, *JATIS*, 11(4).
3. Richer, H. B., Caiazzo, I., Du, H., Grondin, S., Hegarty, J., Heyl, J., Kerr, R., **Miller, D. R.**, and Thiele, S. Massive White Dwarfs in Young Star Clusters. 2021, *ApJ*, 912:165. (Author order alphabetical after the first two)
4. Tzanavaris, P., Gallagher, S. C., Ali, S., **Miller, D. R.**, Puntinga, S., and Johnson, K. E. Cosmic Pathways for Compact Groups in the Milli-Millennium Simulation. 2019, *ApJ*, 871:242.

Unpublished and In Preparation

1. **Miller, D. R.**, Yan, H., et al. Characterization of Candidate High-Mass White Dwarf Escapees Candidates from Open Clusters. In prep.
2. Yan, H., **Miller, D. R.**, Li, C., et al. Photometric and Spectroscopic Analysis of an Extremely Low-Mass White Dwarf Binary. In prep.
3. **Miller, D. R.**, Caiazzo, I., and Heyl, J. Atypical White Dwarfs in Open Clusters. In prep.
4. **Miller, D. R.**, Heyl, J. A Spectroscopic Census of Cluster White Dwarfs from DESI, LAMOST, and SDSS. In prep.
5. Heyl, J., Caiazzo, I., Richer, H., and **Miller, D. R.**. Reconstructing Nearby Young Clusters with Gaia EDR3. *arXiv pre-prints* arXiv:2110.04296.

Observing Programs as PI

Principle investigator on 15 successful observing programs with Gemini, CFHT, and HST, with a strong record of securing competitive telescope time

Gemini	<i>Spectroscopic confirmation of escaped high-mass white dwarfs for the cluster initial-final mass relation</i>	2026
Gemini	<i>Expanding the high-mass end of the cluster white dwarf initial-final mass relation</i>	2026
HST (Co-PI)	<i>Hidden in Plain Sight? Unraveling the Potential Merger Histories of Ultra-Massive White Dwarfs in the Initial-Final Mass Relation</i>	2026
Gemini	<i>Tracing contamination and stellar interactions in the white dwarf populations of open clusters and their tidal tails</i>	2026
CFHT	<i>Probing the mass gap in the white dwarf initial-final mass relation with the rich intermediate-aged open cluster NGC 2360</i>	2026
Gemini	<i>Probing Accelerated Cooling and Formation in a White Dwarf Binary from a Young Open Cluster</i>	2026

CFHT	<i>Probing the white dwarf initial-final mass relation mass gap in intermediate age open clusters with CFHT MegaCam</i>	2025
Gemini	<i>Probing Non-Monotonic Trends in the Open Cluster White Dwarf Initial-Final Mass Relation</i>	2025
Gemini	<i>Developing the White Dwarf Initial-Final Mass Relation by Advancing the Survey of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>Improving the Census of High-Mass White Dwarfs in Open Star Clusters</i>	2024
Gemini	<i>High Mass White Dwarf Escapees From Clusters</i>	2023
Gemini	<i>An Extremely Massive White Dwarf Member of the vdB-Hagen 99 Cluster</i>	2022
Gemini	<i>An Ultramassive White Dwarf Escaped From the Hyades Cluster</i>	2021
Gemini	<i>Reconstructing Nearby Young Clusters with Gaia EDR3: Two White Dwarfs Escaped from Alpha Per</i>	2021
Gemini	<i>Spectroscopy of Massive White Dwarfs in Extended Survey Around Young Clusters</i>	2021

Select Talks and Presentations

Klosterneuburg, Austria	24th European White Dwarf Workshop , ISTA	Contributed Talk	2026
Halifax, Canada	CASCA (Canadian Astronomical Society) 2025	Contributed Talk	2025
Burnaby, Canada	Cosmology Seminar , Simon Fraser University	Invited Talk	2025
Barcelona, Spain	23rd European White Dwarf Workshop	Contributed Talk	2024
Padova, Italy	EAS (European Astronomical Society) 2024	e-Poster	2024
Toronto, Canada	CASCA 2024: Harvey Richer Special Memorial Session	Invited Talk	2024
Vancouver, Canada	Astronomy Seminar , University of British Columbia	Invited Talk	2022
Tübingen, Germany	22nd European White Dwarf Workshop	Contributed Talk	2022

Teaching

Note: After 2023, I did not take on additional TA assignments because my graduate fellowship removed the requirement and my supervisor wanted me to prioritize research. In 2026, I was specifically asked to TA the biennial graduate Stellar Astronomy course, reflecting my standing as the department's leading graduate student in stellar astrophysics.

Astro 508: Stellar Astronomy , TA	2026
Sole TA for this graduate course. Duties include holding office hours, supporting student learning, and managing grading.	
Astro 310: The Solar System , Head Tutorial TA	2022–2023
Oversaw tutorials, managed the course gradebook, held office hours, and revised tutorial materials now used in subsequent offerings.	
Astro 311: Stars and Galaxies , Tutorial TA	2023
Astro 102: Intro to Stars and Galaxies , Tutorial TA	2022
Phys 131: Energy and Waves , Tutorial TA	2021
Astro 101: Introduction to the Solar System , Tutorial TA	2020–2021
Ran and supported tutorials, held office hours, and marked coursework.	

Academic Service & Leadership

JOURNAL REFEREE – <i>ApJ</i>	2026–present
VOLUNTEER – Testing Gravity, Vancouver, BC	2019, 2025

SOLE ORGANIZER – Compact Object Journal Club, *University of British Columbia* 2021–2024

PROBLEM SETTER AND JUDGE – High-Energy Astrophysics Olympiad, *University of British Columbia* 2022

Outreach

PUBLIC TALKS

Space Camp Saturdays, *H. R. MacMillan Space Centre, Vancouver, BC* 2026

Science Rendezvous, *University of British Columbia* 2026

PUBLIC ENGAGEMENT

Astronomy on Tap, *University of British Columbia* 2024–present

Astronomy on Tap, *University of Toronto* 2019

Cronyn Observatory, *University of Western Ontario* 2018

Research Affiliations & Professional Development

Legacy Survey of Space and Time (LSST), *Canada* 2025–present

- Provisional international data rights holder through Canada's in-kind contribution to the Rubin Observatory.

Summer School in Statistics for Astronomers, *Penn State* 2026

- Training in statistical inference, regression and model selection, resampling, Bayesian analysis, MCMC and nested sampling, time series analysis, spatial statistics, and machine learning in R.

ML4Astro Workshop – ICML 2025, *Vancouver, BC* 2025

- Focused on modern machine-learning methods in astrophysics, including foundation models, simulation-based inference, anomaly detection, and AI-assisted scientific workflows.

Testing Gravity Pre-Conference School, *Vancouver, BC* 2025

- Advanced review of gravitation and cosmology in preparation for the Testing Gravity conference.

CRAQ Summer School on *Space Astronomy*, *Montreal, QC* 2023

- Training in space-based astronomy, satellite operations, and the development of astronomical missions and instrumentation.

CASTOR Science Team - Stellar Astrophysics Working Group, *Canada* 2022–2023

- Contributed to development of the stellar astrophysics science case for CASTOR, a proposed Canadian-led UV-optical space telescope.

Testing Gravity Pre-Conference School, *Vancouver, BC* 2019

- Introductory training in gravitation and cosmology in preparation for the Testing Gravity conference.

CRAQ Summer School on *Large-Scale Astrophysics*, *Montreal, QC* 2018

- Training in galaxy formation and evolution, galaxy clusters, and the use of large-scale structure as a cosmological probe.